Numerical Investigation of Pyrolysis Gas Interaction with Hypersonic Reentry Flow-Field

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Abstract

Charring ablative materials used in a thermal protection system (TPS) of atmospheric reentry vehicle systematically eliminates heat from the material through a thermo-chemical process. It involves endothermic plastic decomposition of virgin material into porous char with the production of low molecular weight gas called pyrolysis gas. This gas removes heat by transpiration cooling as it percolates through the porous char material. The gas that escapes from the material forms a thin layer between the fluid and solid domain, thickening the boundary layer, altering the interface properties, and protecting the material surface from the external heat loads. Therefore, it affects the overall design of a TPS system. This work attempts to study the effect of introducing pyrolysis gas from a material surface and its influence on flow and surface interface properties. Flow simulations in this study are performed using open-source direct simulation Monte Carlo (DSMC) solver named Stochastic PArallel Rarefied-gas Time-accurate Analyzer (SPARTA), which is capable of solving multispecies gas flows. A parametric study of the effect of various surface jet parameters on flow and surface interface properties is carried out in a systematic manner.

Motivation

- In atmospheric reentry vehicle, conjugate flow-thermal analysis involves coupling fluid and solid domain at the common interface.
- In charring ablative TPS, pyrolysis gas emitted from solid surface interacts with hypersonic flow-field and influences the flow & surface properties.
- Understanding the pyrolysis gas interaction with external flow-field is crucial in vehicle heat shield design

Objectives

- To study the interaction of pyrolysis gas hypersonic freestream by simulating the physical situation using DSMC technique.
- In DSMC, pyrolysis gas is introduced as jets from the surface elements that forms the vehicle geometry.
- To study the effect of pyrolysis jet parameters such as number density, temperature and velocity on the flow & surface properties.

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Case study

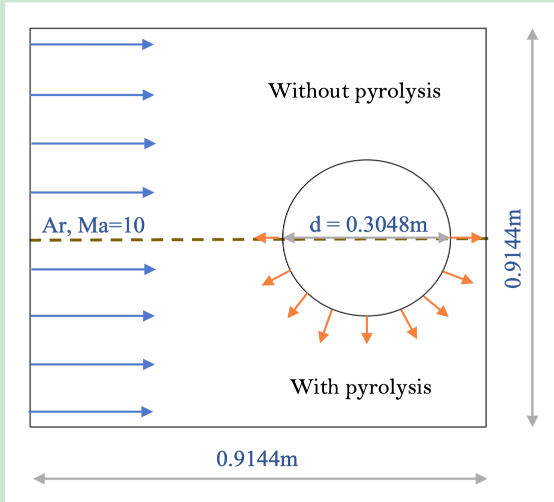

Figure 1: Schematic for base case

Parameter	Freestream	Pyrolysis gas
No. density	$1.7X10^{19} m^{-3}$	$1.7X10^{19} m^{-3}$
Velocity	2634.1 ms^{-1}	10 ms^{-1}
Temperature	200 K	500 K
Kn	0.25	0.25

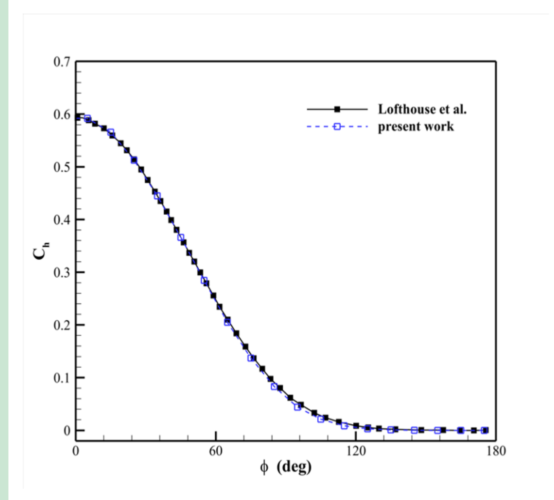

Figure 2: Validation against Lofthouse et al[1]

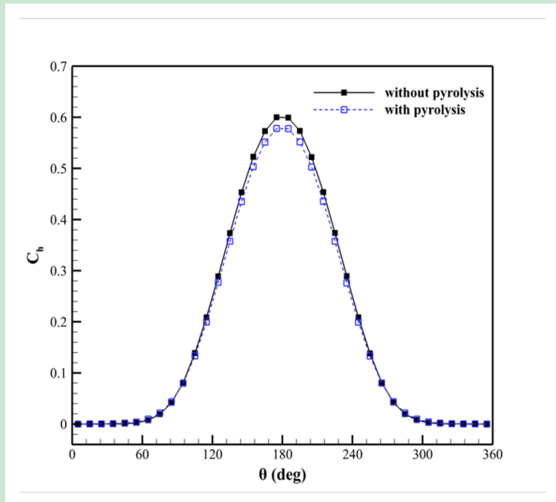

Figure 3: Comparison of non-dimensional heat flux with and without pyrolysis gas.

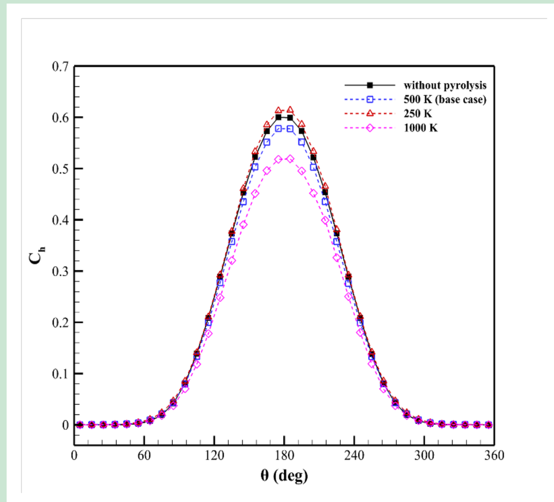

Figure 4: Varying pyrolysis gas temperature

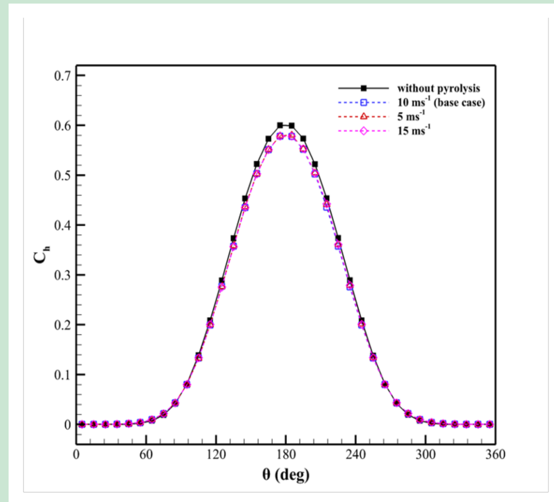

Figure 5: Varying pyrolysis gas velocity.

Results

Comparison of flow properties (top) and stagnation line properties (bottom) for the highest heat flux reduction case.

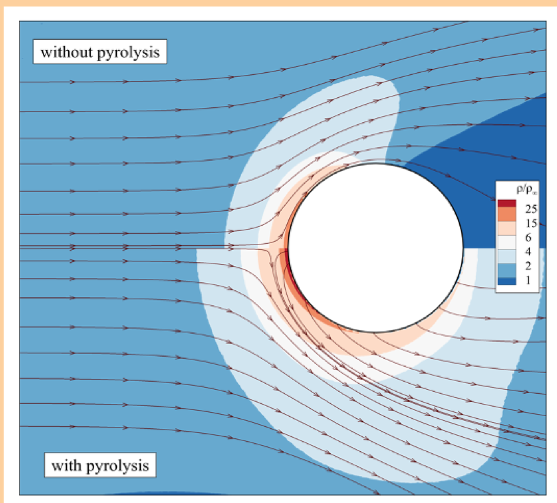

Figure 7: Density contour

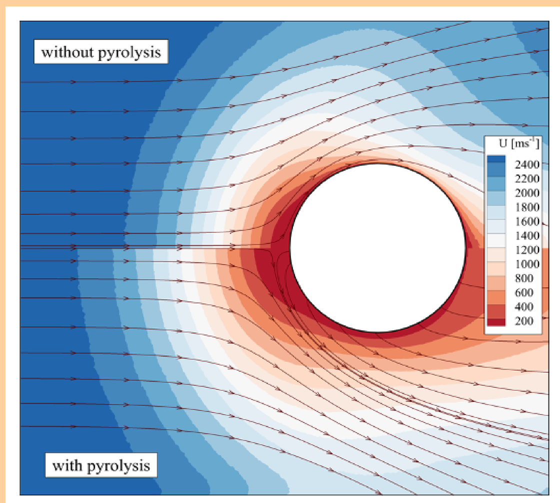

Figure 8: Velocity(x) contour

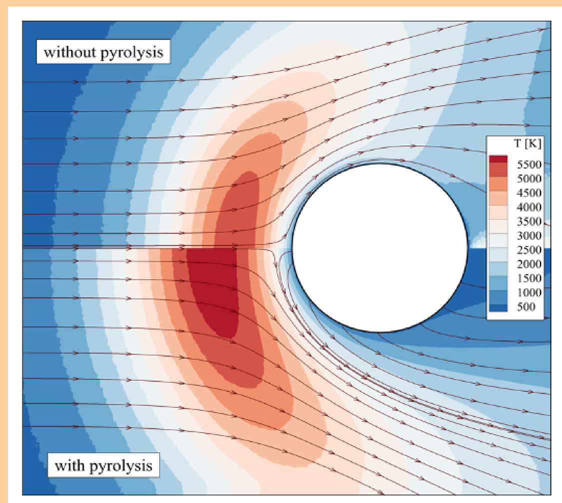

Figure 9: Temperature contour

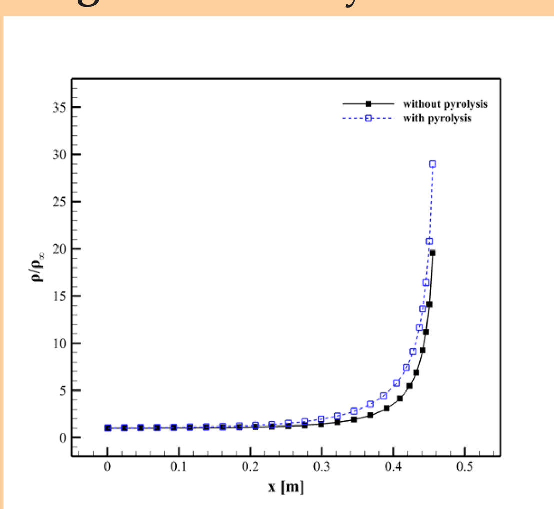

Figure 10: Density variation

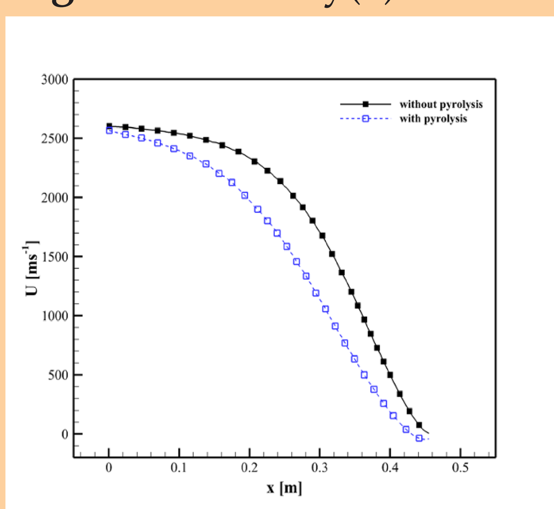

Figure 11: Velocity variation

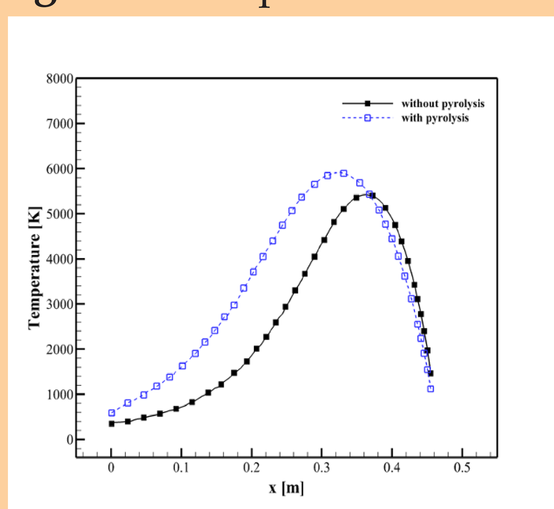

Figure 12: Temperature variation

References

[1] Andrew J Lofthouse, Iain D Boyd, and Michael J Wright. Effects of continuum breakdown on hypersonic aerothermodynamics. *Physics of Fluids*, 19(2):027105, 2007.

Comparison

Varied parameter	Value	HF reduction
No. density (higher)	$1.699X10^{20} m^{-3}$	31.9%
No. density (lower)	$1.699X10^{18} m^{-3}$	-3.48%
Temperature (higher)	1000 K	10.3%
Temperature (lower)	250 K	-6.1%
Velocity (higher)	15 m/s	-0.2%
Velocity (lower)	5 m/s	-0.04%

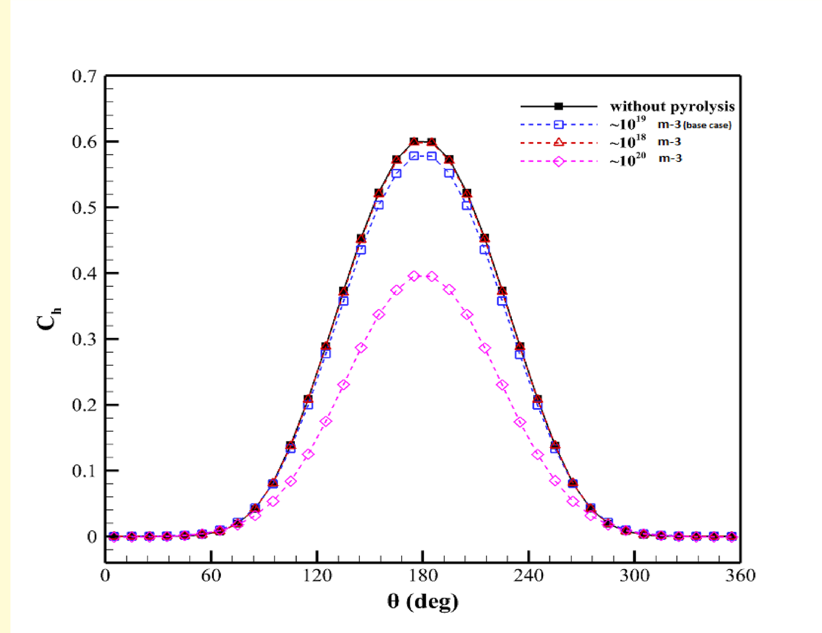

Figure 6: Varying number densities of pyrolysis gas

Further analysis has been done for higher no. density (10^{20}) case because it gives max. heat flux reduction.

Conclusions

- Introduction of pyrolysis gas was found to affect the flow and surface properties.
- Maximum heat flux reduction is experienced when pyrolysis number density is higher than freestream number density. Shock stand-off distance also increases in this case.
- Higher temperature of pyrolysis gas leads to a reduction in the heat flux.
- Velocity has relatively negligible effect on the heat flux in vehicle heat shield design.

Future work

- The numerical investigation can be performed for Knudsen number corresponding to near continuum and transitional regimes.
- The study can be extended to diatomic gas.
- Chemical reaction between pyrolysis gas and freestream gas and its influence on the heat flux can be understood.
- The ablated material coming off the surface will act as granular particles and a combined gas-granular flow study can be performed to understand this.